

MUZZAMMIL SARWAR

Senior Full-Stack Engineer | TypeScript · Node.js / NestJS · Next.js · AI Integration

Islamabad, Pakistan | +92 335 9348075 | muzzammil@umaish.com

linkedin.com/in/muzzammil-sarwar | github.com/Muzzammil242 | muzzammil242.github.io

SUMMARY

Full-stack engineer with 5 years of experience, currently the sole engineer behind a production workforce-management platform of ~130,000 lines of TypeScript: five role-based web portals, a NestJS API, a desktop agent running on a live fleet (Windows, with macOS and Linux ports), a multi-currency finance platform, and the infrastructure and CI/CD behind them. AI features in production: retrieval-augmented matching on pgvector, LLM document parsing and vision-based screenshot classification. Uses AI coding agents (Claude Code) as a force multiplier, with tests and review as the gate. Known for root-causing production issues and fixing the class of bug, not just the instance. Open to remote roles.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Bash

Backend: Node.js, NestJS, Express, FastAPI, REST APIs, BullMQ, Redis, Auth.js / JWT, Webhooks

Frontend: Next.js (App Router, Server Components), React, Tailwind CSS, Radix UI, React Hook Form, Zod

Data: PostgreSQL, Prisma, pgvector (HNSW), Query Optimisation & Indexing, Versioned Migrations, ETL & Reconciliation

AI / LLM: OpenAI API, Google Gemini (Vision), RAG & Embeddings, Structured Outputs, LangChain, CrewAI

Desktop: Electron, Auto-Update Pipelines, Native Modules & FFI, Windows UI Automation (C#), macOS & Linux Builds

DevOps: Docker, Docker Compose, GitHub Actions, Caddy, Cloudflare (DNS, R2, WAF), Tailscale, Linux

Testing: Jest, React Testing Library, Test Automation

AI-Assisted Engineering: Claude Code, Cursor (agentic workflows, custom skills, project memory; test- and review-gated output)

EXPERIENCE

Full-Stack Developer | Growth Rise Solutions

May 2026 – Present

Remote · Sole engineer on Controla-VA, a workforce time-tracking and productivity platform for distributed teams

Platform & architecture

- Built the platform end to end (~130,000 lines of TypeScript): five role-based Next.js portals (admin, client, time tracker, recruiting, finance) on a NestJS API with 30+ modules, PostgreSQL and Prisma, with office-scoped role-based access control. A 55,000-line test suite gates every deploy.
- Led a three-tier refactor that moved all data access and business logic behind the API and replaced two monolithic client shells (2,800+ and 1,600+ lines) with App Router routes.
- Built rule-based anomaly alerts with a manager review workflow, and an office-scoped device block list keyed to hardware identity.

Desktop agent (Windows, macOS, Linux)

- Built the Electron agent running on ~50 workstations across Asia and Latin America: activity sampling, multi-monitor screenshots, idle detection, an offline queue with idempotent sync, in-app task management, and silent auto-updates from a versioned release channel with a server-enforced minimum version.
- Added native Windows integration (C# UI Automation via FFI) that captures the active website for ~95% of browsing time, including in antideetect browsers.
- Porting the agent to macOS and Linux behind a build-time platform layer that strips other operating systems' code: macOS DMG built in CI (pending Apple signing); Linux on X11 first, then Wayland via a GNOME Shell extension, shipped as a self-extracting installer with its own updater.

Reliability & performance

- Traced an over-billing defect (a 17-minute session recorded as 16.5 hours) to open sessions billed against wall-clock time; fixed it server-side with an evidence-based clamp and a session reaper, correcting every installed agent version without a client update, then closed a second path where late offline evidence re-billed dead gaps.
- Designed an event-sourced billing model in which billed time is the union of verified intervals, making over- and double-billing impossible by construction; running in shadow mode against live billing.
- Fixed night shifts vanishing from daily and monthly totals by clipping sessions at midnight on each worker's own time zone (DST-safe), for teams spread from Guatemala to Karachi.

- Traced false tamper and clock-skew alerts to unclear power loss using Windows crash events, and anchored the agent's trusted clock to the OS boot ID on all three platforms.
- Ran a measured code-quality and scalability audit; replaced a per-minute full-table scan (1.17 billion rows read to date) with a partial index.

AI features

- Built retrieval-augmented candidate matching: CVs are embedded once into pgvector (HNSW), each job retrieves its nearest candidates, and only those are scored by an LLM, so cost stays flat as the database grows.
- Shipped Gemini vision classification of screenshots (productive vs. distracting), gated on app and URL usage data so the model only sees ambiguous cases.
- Built LLM-based invoice and CV parsing with a hybrid fallback that handles scanned and malformed PDFs.

Finance platform

- Built a finance consolidation platform (~37,000 lines) that imports and reconciles multi-currency statements from Israeli banks (Hebrew exports), PayPal, Payoneer, Airwallex and eBay, matches payments to invoices, handles VAT and exact-precision FX, and replaces a manual spreadsheet month-end.
- Built a three-tier P&L dashboard (gross, operating and company profit) and a grounded AI assistant that answers finance questions from a bounded ledger snapshot via structured outputs; in acceptance testing with the finance team.

Integrations & infrastructure

- Integrated Zoom: embedded Meeting SDK, webhooks, automatic cloud recording to R2, transcripts and AI summaries.
- Built a WhatsApp recruiting funnel: GoHighLevel automation workflows driven by a webhook event bridge from the candidate landing page.
- Run production on Docker, Caddy and Cloudflare with separate staging and production environments, GitHub Actions CI/CD over a Tailscale tunnel, nightly database backups and versioned Prisma migrations.
- Built an audit log that records the real client IP across a four-hop proxy chain (Cloudflare, Caddy, Next.js, API); the API trusts the forwarded IP only behind a shared-secret internal header, so callers cannot spoof it.

Next on the roadmap

- In-app client-to-worker messaging with automatic Hebrew-Urdu translation of text and voice, and an AI voice interviewer for first-round candidate screening.

Python / AI Engineer | Growth Rise Solutions · Remote

Dec 2024 – Apr 2026

- Built multi-agent AI systems with CrewAI and LangChain (custom memory, tool orchestration), connected to CRMs and messaging services through n8n workflows.
- Wrapped AI logic in high-concurrency FastAPI services, deployed in containers through GitHub Actions to Railway with versioned releases and rollback.
- Built a hybrid recommendation microservice (LightFM, collaborative and content-based) for a live grocery app (box.com.pk), plus Next.js and Streamlit front ends for AI products.

Software Engineer | Bitnine Global · Islamabad

Apr 2023 – Dec 2024

- Built a Python test-automation framework for regression, smoke and sanity testing across Bash, PowerShell and CMD.
- Engineered an Oracle-to-PostgreSQL migration tool (CLI and desktop GUI) covering DDL extraction, schema transformation and data migration, designed to extend to DB2 and MySQL.

Full-Stack Developer (MEAN) | Zepto Systems Ltd. · Islamabad

Apr 2021 – Aug 2022

- Built Angular front ends and Express/MySQL REST APIs, with Strapi as a headless CMS.

PROJECTS

Smart Cold Storage Dashboard: full-stack dashboard for cold-storage automation with real-time temperature monitoring and ML forecasting for predictive maintenance and energy efficiency.

Kids Learning Platform: eye-tracking model for real-time engagement analysis plus APIs around a fine-tuned GPT-3 model that personalises learning content; owned front end and back end.

Engineering case studies: muzzammil242.github.io/#case-studies (six architecture write-ups)

EDUCATION

BS Computer Science | National University of Sciences and Technology (NUST), SEecs, Islamabad

2019 – 2023